

A Soft Wearable System for Real-time Respiratory Disease Diagnosis and Vibrotactile Intervention

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Abstract

Wearable technologies have transformed physiological monitoring, yet most remain passive systems that record data without enabling timely intervention. This limitation is particularly significant for chronic and recurrent respiratory diseases, where early pulmonary abnormalities frequently go unrecognized outside clinical settings. Here, we report a physiologically aware, closed-loop, soft wearable platform that integrates on-body acoustic sensing, machine-learning-based respiratory state classification, and vibrotactile actuation to enable real-time detection and guided response. The skin-interfaced device continuously captures cardiorespiratory sounds and identifies clinically relevant abnormalities, including wheezes, crackles, and rhonchi, triggering condition-specific haptic feedback intended to prompt corrective breathing behavior. To train and validate the classification framework, pulmonary acoustic data were collected from a clinical cohort comprising 32 adults with respiratory disease and 8 pediatric patients with persistent asthma, as well as recordings from healthy participants. The system further delivers personalized guided-breathing cues through adaptive vibrotactile patterns that dynamically reflect each user's respiratory profile. By directly coupling automated diagnosis with immediate actuation, this translational strategy moves wearable respiratory monitoring beyond passive observation toward active, patient-guided management. Such closed-loop functionality enables earlier response to physiological

deterioration and establishes a scalable foundation for preventive, remote, and decentralized respiratory care.

Teaser

A soft wearable platform integrates real-time respiratory diagnosis with adaptive vibrotactile feedback

MAIN TEXT

INTRODUCTION

Wearable medical systems have transformed personal healthcare by enabling continuous and real-time monitoring of physiological signals outside traditional clinical settings (1–6). Flexible and skin-integrated electronics have expanded the capabilities of biomedical devices from sporadic measurement to persistent observation of dynamic biological processes (7–9). These developments have enabled real-time tracking of vital parameters such as cardiac rhythm (10–12), body temperature (13, 14), oxygen saturation (15, 16), breathing acoustics (17, 18), muscle activity (19, 20), and other biomechanical signals (21). These technologies have laid the foundation for personalized and preventive healthcare.

Despite this progress, most wearable devices operate primarily as unidirectional sensing platforms, collecting and transmitting data without responding to users in real-time. For many applications, such passivity limits their practical utility. For respiratory health monitoring in particular, early deviations in breathing patterns, adventitious lung sounds, or cardiopulmonary coupling often precede physiological deterioration (22–24). Yet current digital stethoscopes, stationary lung sound monitors, and wearable respiratory belts typically provide retrospective measurements that require later review (25–27). Without immediate, state-aware feedback cues that directly reflect the current respiratory condition, users may overlook early signs of abnormal respiratory events, contributing to delayed intervention and reduced adherence. This challenge is especially relevant for chronic diseases such as asthma or chronic obstructive pulmonary disease, where adherence to prescribed inhaler therapy frequently falls around thirty percent (28). In parallel with sensing technologies, recent progress in haptic and interactive systems has highlighted the functional impact and demonstrated the beneficial effects of two-way interactions (29–31). These systems demonstrate that wearable interfaces can deliver mechanical cues that communicate information through tactile perception. Such interaction offers significant potential for medical applications. Physiological events detected in real time could guide user behavior, encourage self-correction, and support treatment adherence. Despite this potential, fully integrated platforms that combine continuous sensing, real-time classification, and on-skin feedback within a single soft device remain rare. This gap arises in part from engineering challenges related to achieving high-fidelity sensing, efficient data processing, and effective actuation in a compact and comfortable form.

Here, we introduce a fully integrated soft, on-body sensing-actuation system that combines high-fidelity acoustic sensing, machine learning (ML) based real-time classification, and skin-mounted vibrotactile feedback within a single device. The compact wearable unit incorporates a custom acoustic cavity that enhances the capture of cardiopulmonary sounds and achieves a high signal-to-noise ratio (SNR) of 30 dB. Respiratory acoustic signals are continuously recorded, spectrally analyzed, and used to extract physiological parameters, including breathing rate and heart rate. The system continuously streams data to an ensemble ML model that identifies breathing and non-breathing phases and further distinguishes healthy breathing from characteristic respiratory sounds such as wheeze, crackle, and rhonchi in real-time. Based on the current physiological state of the user, the system generates intuitive vibration cues and personalized guided breathing

prompts. The tactile cues are designed to operate within the frequency range of 50 to 150 Hz, which is well perceived by the skin while remaining separated from the acoustic sensing band (200 to 1900 Hz). By adjusting combinations of vibration amplitude, frequency, and duration, the device delivers more than 8 distinct haptic patterns that users can easily differentiate. In addition to reactive alerts, the system can also provide predefined breathing exercises through direct on-skin delivery. Furthermore, it analyzes the breathing patterns of its users and generates individualized guidance that aligns with the inhalation and exhalation characteristics, thereby functioning as a personalized coaching interface. These interactive capabilities transform the device from a passive monitor into an active and responsive platform that enhances real-time awareness and supports adherence. By unifying sensing, diagnosis, and actuation within a single, soft wearable module, our system addresses three major needs in long-term respiratory care. It enables immediate diagnosis instead of delayed analysis. It provides intuitive and actionable on-body feedback rather than simple passive recording. It offers a compact, fully integrated design instead of a collection of separate components. This work establishes a new direction for interactive physiological devices and presents a pathway to improve adherence, support earlier self-intervention, and advance preventive respiratory healthcare.

RESULTS

On-body soft wearable system for respiratory disease diagnosis and active intervention

Real-time physiological sensing is valuable for tracking respiratory status and documenting disease progression, but sensing alone confines wearable systems to a passive role. However, for individuals with chronic respiratory diseases such as asthma or chronic obstructive pulmonary disease, timely intervention is as crucial as monitoring, as early mitigation can prevent episodes from progressing into acute exacerbations (32). Our system addresses this issue by coupling real-time sensing with on-body, intuitive feedback driven by ML-based diagnosis (Fig. 1A). The device captures high-fidelity cardiopulmonary acoustic signals, including lung sounds and heart sounds, within the acoustic frequency range of 150 to 2000 Hz and wirelessly transmits them to a portable hub. The hub, which serves as both receiver and computation interface, displays real-time physiological status and ML-based diagnosis results through an integrated user interface. Based on the diagnostic output, the hub then triggers a command for an active intervention back to the wearable module. Vibrotactile cues, delivered through mechanoreceptor-responsive vibrations in the 50 to 150 Hz range, provide instantaneous and intuitive alerts while remaining decoupled from the acoustic sensing band.

Fig. 1B and 1C show the top and bottom views of the device. The small and lightweight module conforms closely to the curvature of the chest through a flexible and soft damping encapsulation that reduces external vibration noise. The bottom view highlights the mechanical and acoustic structure, which includes an acoustic cavity that improves sensitivity to low-amplitude cardiopulmonary sounds and two compact vibroactuators that generate tactile cues using small but strong electromagnetic actuators. A skin-safe and reusable brown adhesive surface allows repeated attachment without irritation and maintains a stable interface that supports reliable auscultation and consistent delivery of tactile feedback during daily activities. This integrated layout establishes a key capability of the system, which is the ability to collect physiological signals and provide immediate feedback using one wearable unit. These characteristics can be further understood when compared with prior devices, as summarized in the radar plot in Fig. 1D (29, 33–35). From the system perspective, our device achieves a compact and lightweight form factor while maintaining high acoustic sensitivity, supported by the acoustic cavity and soft enclosure that together improve the SNR by up to 30 dB. From the clinical perspective, the system provides information on five cardiopulmonary features, including wheezing, crackle, rhonchi, breathing rate, and heart rate. The machine learning model distinguishes these conditions with an accuracy above 85 percent, enabling reliable real-time interpretation of respiratory states. Based on these diagnostic results, the wearable

module delivers vibration cues whose amplitude, frequency, and duration can be adjusted to produce multiple distinct feedback patterns that users can clearly perceive.

The practical operation of the system is best understood through the detailed use scenario presented in Fig. 1E. The system continuously acquires cardiopulmonary sounds and wirelessly streams them to the portable hub, where a pretrained ML model analyzes the incoming signal in real-time. The model processes the data in two-second windows and updates the classification output every 0.5 seconds through boundary shifting, which enables continuous assessment of respiratory states without interruption. Incoming audio is first classified as either breathing or non-breathing, and segments identified as breathing are further categorized using a multi-class classifier into one of four classes, including healthy, wheeze, crackle, or rhonchi. Each predicted state immediately informs the intervention command delivered back to the wearable unit. Users can visually recognize the alert through an integrated LED, and the simultaneous on-skin vibration cue ensures intuitive perception even when the device is covered by clothing. The vibration pattern is selected according to the identified respiratory class, allowing distinct tactile guidance for different abnormal conditions. Beyond reactive alerts, the system can also initiate predefined breathing patterns that support recovery in respiratory disease management or generate personalized guided breathing cues based on individual respiratory characteristics to help restore a calm and stable breathing or cardiac rhythm.

Design, implementation, and signal acquisition of the soft wearable system

The on-body sensing-actuation integrated system was designed with a multilayer stacked structure that integrates all core components into a single compact module (Fig. 2A and movie S1). At the center, a custom printed-circuit-board (PCB) contains the MEMS microphone, power regulator, MCU, BLE circuit, and haptic driver, forming a fully integrated sensing-processing-actuation circuit (fig. S1). Directly beneath the MEMS microphone is an acoustic cavity that reduces the acoustic impedance mismatch between the skin and the device, improving the capture of cardiopulmonary sounds. The bottommost layer is a reusable, skin-safe adhesive that maintains stable attachment on the body. The actuation module consists of electromagnetic coils connected to the haptic driver and a paired magnet that generate vibrotactile output through current excitation, with a magnified view shown for clarity. All components are powered by a 70-mAh lithium polymer battery and enclosed by a soft elastomeric frame. With its flexible and low-profile structure, the device conforms to both the anterior and posterior chest wall surfaces, ensuring comfortable attachment with minimal perceptual burden to the user (Fig. 2B).

A system-level overview of the device's internal signal flow and communication framework is shown in Fig. 2C. The four functional modules for power, sensing, actuation, and communication operate through dedicated pathways within the integrated circuit. The sensing pathway from device to host streams cardiopulmonary audio captured by the MEMS microphone, processed through an amplifier, and transmitted via BLE using a dedicated Universally Unique Identifier (UUID) assigned to the sensing service. Conversely, the actuation pathway from host to device uses an independent UUID that enables real-time control of the two electromagnetic coils through the haptic driver. These BLE characteristics for sensing and actuation function independently, allowing simultaneous acquisition of auscultation data and delivery of vibrotactile commands. In addition, wirelessly streamed data are processed, visualized, and uploaded to the cloud for remote monitoring and storage. The acoustic air cavity was designed to enhance sound transmission within selected frequency bands. The primary geometric parameters in designing an acoustic cavity include the curvature of the internal wall, the inlet area in contact with the skin, the outlet area facing the MEMS microphone port, and the overall cavity height. The curvature parameter n , which governs the internal wall profile, was selected as $n = 1.5$ based on the curvature-dependent acoustic pressure distribution analysis (fig. S2).

The outlet diameter d_{out} was fixed at 1 mm to match the diameter of the microphone port, while the inlet diameter D and cavity height h were varied to tune the internal volume of the cavity. By adjusting these parameters, specifically h as 3, 4, 5 mm and D as 10, 15, 20 mm, the cavity volume was altered, enabling control of the resonance frequency according to the Helmholtz resonator model. Then, frequency sweep experiments were conducted from 100 to 1600 Hz in 100-Hz increments with each cavity attached to the microphone port (fig. S3A and B). Compared to the reference case without a cavity (black line), all cavity designs showed higher signal amplitude in both the low-frequency cardiac-dominant band (100 to 200 Hz) and the high-frequency lung-sound-dominant band (500 to 1600 Hz). In addition, the structure with a smaller internal volume (pink line) exhibited progressively larger amplitude in the high-frequency range, demonstrating the tunability of the cavity response. These results confirm that the air cavity design improves the signal quality and strengthens specific frequency components of cardiopulmonary sounds. To enhance sensitivity to high-frequency lung sound components, an air cavity with $h = 3$ mm and $D = 10$ mm was selected for the device. In addition, the effect of the cavity was further validated through finite element analysis (FEA), comparing the sound pressure distribution and airflow profile between a simple tubular structure and the designed cavity geometry (fig. S4).

In addition to the acoustic air cavity, the entire structure is enclosed by a soft elastomeric frame that provides mechanical damping, enabling the device to maintain a robust SNR even under motion. This performance was evaluated in comparison with a commercial rigid-type stethoscope (fig. S5). First, the low-profile form factor of our module stands out by achieving a slim fit without protrusion and conforming seamlessly to curved surfaces without gaps. During quiet standstill measurements of relaxed inhalation and exhalation, our device achieved an SNR of approximately 30 dB, which is more than twice that of the comparison device. The benefit of the soft enclosure becomes even more pronounced during movement, where low-frequency body-borne vibrations typically degrade acoustic sensing. While our device effectively attenuates these vibrations and maintains an SNR of about 27 dB during walking, the comparison device shows a marked reduction to around 10 dB. Both the time-domain signal plots and spectrograms clearly illustrate that the rigid-type device exhibits dominant, unfiltered vibration-induced energy spikes, whereas the mechanically damped composition of our device minimizes these artifacts and preserves the clarity of the cardiopulmonary sounds. After acquiring the raw signal, the sensing data undergo a signal processing workflow, with the detailed pipeline described in fig. S6. To separate cardiac and lung sound components, different band-pass filter ranges and maximal overlap discrete wavelet transform reconstruction levels are applied. The refined signals are then converted into a time-domain waveform, a spectrogram, and an energy plot. Using a shifting-window scheme and an adaptive peak-detection algorithm, the system continuously updates breathing rate and heartbeat rate in real-time. This processing pipeline enables the system to distinguish different respiratory conditions, such as healthy normal breath sounds and abnormal breath sounds (fig. S7). In the time-domain waveform and corresponding energy density plot, the inhalation and exhalation profiles are clearly observable. The spectrogram further highlights these differences as healthy subjects show no dominant energy concentration within a specific frequency band, while asthma users exhibit distinct high-intensity traces in the 200–500 Hz range. These features are also evident in Fig. 2D, which summarizes the respiratory and cardiac metrics extracted from an individual with asthma. The processed signal, with noise appropriately filtered, reveals clear inhalation–exhalation profiles, and the spectrogram shows the characteristic wheeze lines near 500 Hz. The combined respiratory and cardiac features enable continuous plotting of breathing rate and heart rate, providing comprehensive real-time monitoring. Furthermore, in addition to healthy and wheezing cases, crackle and rhonchi sounds (fig. S8) were also measured from 32 adult patients, selected according to the clinical study participant demographics, recruitment criteria, and study objectives (table S1), following the clinical acquisition protocol (fig. S9). Each case exhibited characteristic peaks in the

short-time Fourier transform (STFT). Audio examples of the corresponding respiratory sounds for each case are provided in movie S2.

Beyond single-point measurements, our system enables continuous, on-body monitoring of respiratory conditions over clinically meaningful time scales, allowing assessment of treatment progression through changes in breathing patterns and acoustic features. As a representative example, continuous measurements were performed on eight pediatric participants with asthma for durations ranging from 3 to 6 hours, during which respiratory status and treatment response were monitored (patient cohort recruitment criteria and demographics provided in Table S1, and detailed protocol provided in fig. S9 and S10). During hospitalization, patients received pharmacological treatment and were intermittently evaluated by clinicians using the Clinical Respiratory Score (CRS). However, these assessments rely on periodic bedside examinations and therefore provide indirect and time-discrete evaluations of respiratory status. In contrast, as shown in Fig. 2D, the wearable system enables continuous longitudinal monitoring of respiratory sounds between clinical assessments, capturing real-time changes in pulmonary status throughout the admission-to-discharge period. In Fig. 2F, the STFT of the breathing sounds collected over time demonstrates the evolution of respiratory characteristics during treatment. At the initial enrolled state, participants with severe asthma exhibit prominent, narrow-band spectral peaks associated with wheezing. As treatment progresses, these peaks gradually diminish in magnitude, and the final measurements prior to discharge show a transition toward a more uniform spectral distribution characteristic of normal breathing. Such a trend is further supported by the comparison of breathing sounds before and after treatment (fig. S11), where both the overall breathing intensity and the wheeze-indicating spectral peaks are markedly reduced. This observation highlights the capability of our device to support longitudinal monitoring by continuously tracking respiratory abnormalities and their improvement during treatment.

Characterization and real-time integration of the sensing–actuation module.

Our on-body sensing-actuation system employs a compact electromechanical actuator to deliver perceptible physical feedback while maintaining a small and lightweight form factor. To avoid the need for an additional accessory, the actuator is fully integrated within the same device that houses the sensing module, allowing both sensing and actuation to operate from a unified platform. As shown in Fig. 3A, the actuation module consists of a tightly wound fine-gauge copper coil embedded in the device frame and a small permanent magnet with a 3 mm diameter and 1 mm thickness attached to the center of a compliant elastomer membrane. The membrane is fixed at its perimeter, leaving the central magnet free to oscillate in response to alternating current applied to the coil. This configuration enables the magnet to undergo controlled reciprocating vertical motion, generating tactile feedback on the skin (fig. S12). The interactive feedback force transmitted to the skin is determined by the geometric parameters of the coil-magnet assembly, the magnetic properties and size of the magnet, and the amplitude of the driving current (supplementary note 1 and fig. S13). Five coils with different coil geometric parameters were selected to tune the mechanical output while minimizing the electrical energy consumed during actuation (fig. S14). The coil dimensions were determined based on their geometric ratios relative to the permanent magnet, specifically the coil length-to-magnet length ratio and the coil radius-to-magnet ratio. A low-profile wearable unit requires not only an integrated sensing-actuation structure but also a minimized battery size, while at the same time, the capacity needed to power all electrical components.

To enable prolonged operation with a small-scale battery, the actuator design requires careful consideration of energy efficiency. The simulated actuation efficiency derived from these parameters is shown in Fig. 3B and provides a basis for optimizing and fabricating the coil design. As indicated by the analytical model, the actuation efficiency increases as the coil length decreases. However, because the actuator is driven by a voltage-mode driver, excessively shortening the coil

299 reduces its electrical resistance and can lead to elevated Joule heating under a given drive condition.
300 Therefore, to balance energy efficiency with thermal conditions, a coil length of 3 mm was selected
301 for the design. In addition, increasing the coil diameter enhances the force generated at the given
302 scale (Fig. 3C). Considering this force scaling together with the efficiency trend from the analytical
303 model, we selected a coil geometry with a radius ratio of 1.5. Meanwhile, when alternating current
304 is applied to the electromagnetic coil to induce magnet oscillation, resistive Joule heating can lead
305 to a rise in coil temperature. This effect becomes more pronounced at lower driving frequencies,
306 where the actuator spends a longer duration in each current cycle. However, with the coil
307 dimensions, driving frequencies, and current amplitudes used in our system, the temperature
308 remains stable during continuous operation. As shown in Fig. 3D, driving the actuator at frequencies
309 above 50 Hz maintains the temperature below 35 °C even after more than 120 seconds of
310 uninterrupted excitation.

311 The magnetic actuator can generate a wide range of vibrotactile sensations by adjusting key
312 driving parameters, including current amplitude (I_c), driving frequency(f_c), and actuation
313 duration(t_c). Fig. 3E shows representative examples of how variations in these parameters shape the
314 resulting feedback, demonstrating the capability of our system to produce distinct tactile cues. In
315 addition, fig. S15 provides a detailed classification of the vibration patterns that can be produced
316 by systematically varying each parameter. Furthermore, the actuator can generate diverse waveform
317 patterns such as spike, envelope, regression, Fourier superposition, and square waveforms through
318 mathematical modulation of these parameters (supplementary note 2). As shown in Fig. 3F, these
319 parameters enable the construction of diverse actuation profiles, such as homogenous spikes, pulse-
320 like spikes, sequentially increasing or decreasing amplitude patterns, square or periodic triangular
321 waveforms, amplitude-modulated sinusoidal waveforms, and a mixture of waveforms. After
322 establishing the range of achievable actuation profiles, it is essential that these feedback signals be
323 delivered precisely when needed, without interfering with the performance of the sensing module
324 integrated within the same device. As a representative use scenario, our system was attached to the
325 body during daily activities while continuously performing real-time monitoring (Fig. 3G). When
326 an abnormal respiratory event was simulated, a physical alert was subsequently triggered. Because
327 the vibrotactile actuator and the MEMS microphone coexist in physical proximity, the mechanical
328 vibrations used for alerting inevitably appeared in the sensor output. This effect is evident in the
329 middle panel of Fig. 3G, where the actuator-induced vibrations are superimposed on the
330 physiological acoustic signal during both the alert (Cue 1) and the guided-breathing intervention
331 (Cue 2) that follows. To isolate the physiological components of interest, the actuation frequency
332 was intentionally selected to lie outside the respiratory and cardiac bands. As a result, applying
333 band-pass and notch filtering effectively removed the actuation-induced component. The lower
334 panel of Fig. 3G shows the decoupled sensor signal, demonstrating that cardiorespiratory sounds
335 can be preserved and processed during simultaneous sensing and actuation.

337 **Machine learning-enabled real-time diagnostic and feedback**

338 To enable real-time assessment and immediate delivery of tactile feedback, our system incorporates
339 a real-time machine learning-enabled diagnostic algorithm rather than relying on stored or
340 retrospectively processed data. As the feedback must respond to the current state, the model is
341 designed to analyze continuously streamed auscultation signals and generate classification outputs
342 on the fly, forming the basis for instant actuation commands. The development of the respiratory
343 sound multi-class classification model followed a multi-stage pipeline beginning with interval
344 parsing of annotated inhalation and exhalation phases from 30-second recordings (Fig. 4A). These
345 annotations were cleaned and converted into breathing versus non-breathing intervals. Raw
346 auscultation audio was resampled to 16 kHz and segmented according to these intervals, with each
347 segment assigned one of five labels, including healthy, crackle, rhonchi, wheeze, or non-breathing.

348 To standardize temporal resolution for learning, all segments were converted into 2-second
349 windows using a 0.5-second hop, with padding applied to shorter segments. Each window was
350 transformed into a 64-band log-Mel spectrogram, normalized on a per-sample basis, forming the
351 input tensor for model training. Before training the model, a group-based data-splitting strategy
352 (GroupShuffleSplit) was applied to prevent data leakage by ensuring that recordings from the same
353 participant were not shared across training, validation, and test sets. Then, logistic regression and
354 multilayer perception classifiers were trained on the log-Mel features, and their output probabilities
355 were combined through ensemble averaging. The final model generated multi-class predictions
356 across the five respiratory states, and each diagnosis triggered the corresponding tactile feedback
357 pattern on the wearable unit. Here, a total of 38 patients were included for model training, and the
358 patient information used is summarized in Tables S1 and S2. Data acquisition was performed using
359 a custom-designed user interface that manages real-time data streaming from the wearable device
360 (movie S3), cloud-based data storage, synchronization, and access (fig. S16).

361 Although the model was designed to minimize computational load for real-time diagnosis,
362 both linear and nonlinear decision boundaries were incorporated through an ensemble of logistic
363 regression and a multilayer perception classifier. This combination allowed the system to capture
364 simple amplitude- or energy-based distinctions with the linear model while simultaneously
365 recognizing more complex and nonlinear spectral patterns with the multilayer perception. As shown
366 in the confusion matrix in Fig. 4B, the final ensemble achieves strong agreement between predicted
367 and actual labels, with classification accuracies of at least 85 % across all four respiratory classes
368 of healthy, wheeze, crackle, and rhonchi, demonstrating reliable discrimination of diverse
369 respiratory states for real-time operation. A representative example of real-time inference from a
370 participant with crackle-dominant breathing sounds is shown in Fig. 4C, where the system
371 processes streamed audio and continuously generates diagnostic outputs. As the incoming
372 waveform arrives at the portable hub, it is divided into 2-second analysis windows that update every
373 0.5 seconds, like shaded blue, orange, and green windows. For each window, the model performs a
374 five-class prediction. Breathing versus non-breathing activity is derived directly from these outputs
375 by grouping the four respiratory classes as breathing and treating non-breathing as a separate
376 category, as shown by the red and navy lines in the middle panel. The shaded red and green intervals
377 correspond to breathing and non-breathing phases, respectively, and the transition points between
378 these phases align with the crossings of the red and navy prediction curves, confirming that the
379 model accurately separates the two phases. The bottom panel visualizes the multi-class probabilities
380 for the four breathing-related states, where the crackle class consistently exhibits the highest
381 probability, matching the ground-truth respiratory pattern of the participant.

382 On the basis of these real-time classification results, Fig. 4D outlines how the system
383 translates diagnostic results into an alert and actuation protocol. Once each 2-second window is
384 evaluated, the predicted respiratory state is immediately used to determine whether an alert should
385 be triggered. Alerts can be initiated by detecting an abnormal respiratory class from the model
386 output or by identifying deviations in HR or BR that exceed predefined thresholds. When an alert
387 condition is met, the system selects the corresponding actuation pattern and delivers it to the
388 wearable unit. As a representative demonstration, Fig. 4E to G shows the complete sequence from
389 real-time diagnosis to alert generation for participants exhibiting wheezing, crackle, or rhonchi
390 (movie S4). Each panel represents the waveform, spectrogram, model-derived probabilities, and
391 resultant tactile feedback. The acoustic patterns of the three conditions are clearly distinguishable
392 across the time-frequency domain, spectrogram, and model output. Notably, the timing of the model
393 predictions aligns well with the inhalation-exhalation cycles. Moreover, the dominant predicted
394 class rises well above the others, with a confidence level exceeding 70 % in all cases. Based on this
395 diagnosis, the system provides the corresponding predefined actuation pattern as shown in the
396 bottommost plots. Each actuation profile reflects the characteristic features of its respective acoustic
397 abnormality. Wheezing is rendered as a sustained amplitude-modulated vibration mirroring its

narrowband tonal structure. Crackles are represented by a sequence of sharp spike-like pulses corresponding to their transient, burst-type signatures. Rhonchi are translated into slow, low-frequency modulated vibrations, capturing their coarse, rumbling nature.

To assess the perceptual discriminability of the vibrotactile actuation patterns, we conducted a user perception test in healthy participants. Participants wore the device and were presented with eight different vibrotactile patterns corresponding to distinct alert conditions, including tactile representations of abnormal respiratory states (wheezing, crackle, and rhonchi) as well as patterns encoding variations in vibration amplitude and temporal modulation. After experiencing each stimulus, participants were asked to identify the perceived pattern from the predefined set. The classification results from all trials are summarized as a confusion matrix in fig. S17. Overall, participants demonstrated a high ability to discriminate between the different vibrotactile patterns, with recognition accuracies exceeding 85 % for all classes. In particular, the tactile alerts corresponding to respiratory abnormalities including wheezing, crackle, and rhonchi were reliably recognized, indicating that the actuation patterns effectively convey distinct diagnostic information. These results confirm that the designed vibrotactile signals can serve as an intuitive feedback interface for communicating diagnostic outcomes generated by the real-time respiratory classification system.

Application as a personalized guidance and care unit

Beyond continuous sensing, the integration of tactile actuation in our system enables the device to actively interact with the wearer and provide physiological guidance. Vibrotactile stimulation can serve as an intuitive communication channel, allowing physiological information to be translated into tactile cues that users can perceive without relying on visual or auditory feedback. Such tactile guidance is particularly attractive for wearable healthcare systems, where discreet and continuous interaction with the user is desirable. One application of this capability is the support of structured respiratory exercises that are commonly used in pulmonary training and breathing regulation (36, 37). These exercises prescribe a specific sequence of inhalation, breath holding, exhalation, and holding intervals, which typically require external coaching or visual guidance. To evaluate whether vibrotactile cues can provide such guidance, we performed a box-breathing exercise in which participants followed a predefined tactile breathing sequence generated by the device (Fig. 5A). In this routine, the vibration envelope encoded the timing of each 4 seconds of breathing phase (inhale–hold–exhale–hold), enabling users to intuitively follow the breathing cycle through tactile cues alone. The plots in Fig. 5A show the vibrotactile patterns delivered and the respiratory signal measured while the participant followed the tactile guidance, demonstrating that the delivered vibrotactile pattern effectively guided the breathing exercise without the need for visual or auditory instructions.

In addition to guiding predefined breathing routines, the sensing capability of the device enables the extraction of individual physiological rhythms, which can be translated into personalized vibrotactile patterns. Prior studies have shown that rhythmic mechanical stimulation can promote physiological relaxation and synchronization with external cues (38–42). Building on these findings, the ability of our system to reproduce an individual's baseline breathing and cardiac rhythms through tactile signals provides a framework for personalized physiological guidance, where familiar cardiorespiratory patterns may be delivered through vibrotactile stimulation as reference rhythms during periods of stress. During a calm resting period, respiratory sounds recorded by the device were analyzed to derive a representative breathing envelope (Fig. 5B). The respiratory cycle was segmented into inhalation and exhalation phases, after which the corresponding waveforms were averaged to obtain a characteristic breathing profile (Fig. 5C). From this profile, key parameters including inhalation amplitude and duration (A_i , t_i) and exhalation amplitude and duration (A_e , t_e) were extracted together with a fitted breathing envelope.

The extraction of breathing sound features and the derivation of the fitted curve are described in detail in Supplementary Note 3 and fig. S18. Using this information, the system can generate a vibrotactile pattern that reproduces the extracted breathing dynamics (Fig. 5D). Fig. 5B–D shows a representative example from a participant, while the consistency of breathing-pattern extraction and tactile reconstruction across healthy participants is summarized in fig. S19. A similar approach can be applied to cardiac rhythms. Heart sounds recorded during a resting baseline period were analyzed to extract the characteristic timing and relative amplitudes of the S1 and S2 components of the cardiac cycle (Fig. 5E–G). Results from participants demonstrating consistent cardiac-pattern extraction are summarized in fig. S20. These parameters were then used to generate a vibrotactile pattern that reproduces the individual heartbeat envelope (Fig. 5H and movie S5). Such extracted individualized physiological rhythms can therefore serve as personalized reference cues for vibrotactile stimulation, facilitating physiological relaxation through mechanical stimulation.

Taken together, these results demonstrate that the integrated sensor–actuator architecture enables the device to move beyond passive physiological monitoring toward active and interactive physiological guidance. By combining continuous sensing with tactile actuation, the wearable system can not only capture the wearer's current physiological state but also translate this information into vibrotactile cues. This capability allows the device to provide intuitive feedback based on sensed physiological dynamics, guide structured respiratory exercises, and reconstruct individual cardiorespiratory rhythms through tactile signals. Such sensor–actuator integrated systems therefore serve as a platform for interactive healthcare applications, where physiological monitoring, active intervention, and guided feedback are seamlessly combined within a single wearable interface.

DISCUSSION

The soft, closed-loop wearable system presented in this work establishes a fully integrated, on-body sensing–actuation platform that combines high-fidelity cardiopulmonary acoustic sensing, real-time ML-based diagnosis, and on-skin vibrotactile actuation within a single soft, compact module. Through a custom acoustic cavity and soft elastomeric housing, the device achieves robust acquisition of low-amplitude respiratory and cardiac signals, while the lightweight ensemble ML model enables continuous classification of healthy and adventitious lung sounds with high accuracy. Coupled with an energy-efficient dual-actuator module, the system delivers intuitive tactile alerts and personalized guided-breathing cues, demonstrating the potential of interactive wearables to support adherence, promote self-regulation, and enable decentralized respiratory cues. As summarized in Table S3 (17, 43–45, 35, 46–48), our system further distinguishes itself from existing wearable pulmonary monitoring systems by offering high SNR, multi-feature cardiopulmonary sensing, and real-time ML-based diagnostic capability that moves beyond passive data collection. Most importantly, the incorporation of bi-directional interaction positions our system as a new class of interactive medical interface devices. There remain several promising opportunities to extend the capabilities of our system. Although the current real-time classification pipeline operates on fixed two-second windows for continuous inference, future implementations may incorporate adaptive segmentation models that dynamically detect inhalation and exhalation boundaries (49). Furthermore, the use of pretrained audio models combined with transfer learning will offer a pathway to exploit knowledge from large-scale heterogeneous datasets, helping to overcome the inherent data scarcity and noise present in medical care settings (50, 51). While this work focuses on single-site cardiopulmonary sensing, integrating multi-site arrays may allow more comprehensive characterization of cardiorespiratory mechanics and improve robustness to user-specific variability (52). Lastly, large-scale longitudinal studies across diverse clinical populations

will validate long-term stability, optimize personalization algorithms, and evaluate the impact of the system on self-management and therapeutic adherence in chronic respiratory diseases.

MATERIALS AND METHODS

Human subject study. Human subject testing was conducted on both healthy individuals and symptomatic patients to collect respiratory acoustic data for pulmonary disease detection and model development. A total of 43 participants were enrolled, including 5 healthy adults and 38 symptomatic patients. The symptomatic cohort comprised 32 adult patients with respiratory disease and 8 pediatric patients diagnosed with persistent asthma. Adult patients were recruited and evaluated at Chungnam National University Hospital (CNUH) under Institutional Review Board approval (IRB No. 2025-05-031), while pediatric patients were recruited at Children's Healthcare of Atlanta (CHOA) under Institutional Review Board approval (IRB No. 2025P009892). For both clinical sites, the wearable devices were prepared and supplied by the Georgia Institute of Technology research team. All participants, or legal guardians for minors, provided written informed consent prior to participation in accordance with local IRB regulations. The study consisted of both cross-sectional and longitudinal cohorts. (table S1) The cross-sectional cohort included 32 adult patients with respiratory disease and 5 healthy participants, who completed a single recording session to capture a broad range of respiratory sound patterns. During each session, pulmonary auscultation was performed using both the soft wearable system and a traditional stethoscope at the same chest location, with each recording lasting approximately 3 minutes. Clinical diagnoses and respiratory sound labels were determined by pulmonologists and used as ground truth for machine learning model development. The longitudinal cohort included 8 pediatric asthma patients, who were continuously monitored for a minimum of 3 hours from hospital admission through discharge to capture temporal changes in respiratory status and treatment response. During this period, repeated recordings were obtained using the wearable system, and clinicians documented respiratory severity using clinical respiratory scoring and diagnostic assessment. These longitudinal data were used to evaluate the feasibility of continuous monitoring and to support model development for disease progression tracking. All collected data were stored in a secure cloud database and used for respiratory sound classification model training and validation, as well as longitudinal respiratory progress monitoring.

Device fabrication. The device was fabricated through a sequential multilayer stack-up and molding process. First, the bottom acoustic cavity was formed using a custom-designed 3D printed mold. Polydimethylsiloxane (PDMS; Sylgard 184, DOW) was prepared at a 10:1 base-to-curing-agent ratio, mixed thoroughly, and cured at 65 °C for 4 h. For device integration, the cured PDMS cavity was placed into a second custom 3D printed injection mold designed to define the overall device geometry. A PCB containing the sensing, actuation, and control electronics was aligned and positioned on top of the cavity. MLV Seam Tape (Trademark Soundproofing) was placed on top of the PCB board. After sealing the mold, Ecoflex 00-30 (Smooth-On) was injected into the remaining volume. The mold design incorporated raised features corresponding to the actuator and battery compartments. These regions remained unfilled after casting to create dedicated spaces for later component insertion. Electrical leads for subsequent connection of the battery and coil (JHCOILS) were pre-assembled and fixed in place within the mold prior to casting. Following the injection, the assembly was cured at 65 °C for 30 min and demolded. The battery and electromagnetic coil were then placed into the designated cavities and electrically connected to the PCB using the pre-positioned leads. Additional Ecoflex 00-30 was dispensed to achieve full encapsulation of the device. The electromagnetic coils were custom-fabricated using 40-AWG enamel-coated copper wire wound into an air-core. For the bottom reusable adhesive layer, a silicone gel (High-Tack

Silicone Gel, Factor II) was coated onto a fabric-type medical adhesive (3M 9907T). The adhesive layer was laser-cut to shape and bonded to the underside of the acoustic cavity using Sil-Poxy (Smooth-On). A thin-film Ecoflex 00-30 layer was prepared separately by spin-coating Ecoflex 00-30 at 300 rpm for 30 sec and curing at 65 °C for 15 min. This thin film was laser-cut to create patterned regions defining the attachment sites for both the magnet and the device. A neodymium magnet (3 mm diameter, 1 mm thickness) was placed onto the patterned Ecoflex thin film and bonded using Sil-Poxy, after which the film-magnet assembly was aligned and attached to the surface of the device following the guidelines.

Acoustic performance characterization. The acoustic response of the device was evaluated using a controlled sweep test. A custom soundproof enclosure housing a speaker was used as the excitation source. The top surface of the enclosure was overlaid with a 10 mm thick elastomer layer, and all measurements were conducted with the device adhered directly onto this elastomer surface, positioned directly above the speaker. Devices with different cavity structures were tested under identical conditions. For each configuration, the speaker generated a frequency sweep from 100 Hz to 1600 Hz, played as discrete 5-second segments. A 5 sec silent interval was inserted between segments.

Actuator output force. The output force of the electromagnetic actuator was quantified using a motorized force testing system (ESM303, Mark-10) configured in a compression-testing setup. Devices incorporating coils of different geometries were mounted on the lower stage of the tester. During testing, electrical current was supplied to the coil while the probe maintained a fixed displacement constraint. The repulsive force generated by the actuated coil-magnet was recorded by the load-cell. For each coil design, force-current characteristics were obtained by incrementally increasing the input current and measuring the corresponding steady-state reaction force.

Thermal characterization. The thermal behavior of the actuator coil under varying electrical driving conditions was evaluated using an infrared thermal camera (FLIR). All measurements were conducted in air at a room temperature of 26 °C.

Simulation. The acoustic response of the cavity structures was analyzed using a 2D axisymmetric pressure-acoustic model in COMSOL Multiphysics (v6.2). The geometry consisted of an air-filled cavity defined by a parameterized curved profile, bounded by rigid walls, and driven by a harmonic pressure source at the bottom boundary. Material properties for air, including density and speed of sound, were assigned from built-in interpolation tables. Frequency-domain studies were performed from 100 to 1600 Hz.

Acoustic signal processing. All acoustic signals were processed using a standardized multistage pipeline consisting of band-pass filtering, multiresolution wavelet decomposition, level-specific enhancement, and coefficient-domain denoising (fig. S5). MATLAB was used for the signal processing. For respiratory recordings, raw signals were first band-pass filtered between 150 and 1900 Hz. For cardiac recordings, a 100 to 250 Hz band was used. After the band-pass filter, signals were decomposed into multiresolution sub-bands using MODWT with the sym4 mother wavelet. For each level j , the wavelet and scaling coefficients were obtained as

$$W_{j,k} = (x * \overline{h_j})[k], V_{j,k} = (x * \overline{g_j})[k]$$

where $x(t)$ is the bandpass-filtered acoustic signal, $\overline{h_j}$ and $\overline{g_j}$ denote the level j MODWT wavelet and scaling filters, and k index time.

To further suppress nonstationary noise while preserving transient acoustic features, a 1D Kalman filter was applied to each selected wavelet level ($W_{j,k}$). The recursive estimator followed

$$\hat{x}_{k|k-1} = \hat{x}_{k-1}, P_{k|k-1} = P_{k-1} + Q$$

$$K_k = \frac{P_{k|k-1}}{P_{k|k-1} + R}$$

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k(z_k - \hat{x}_{k|k-1})$$

$$P_k = (1 - K_k) P_{k|k-1}$$

Where Q represents the process noise variance, R is the measurement noise, $\hat{x}_k$ is the denoised coefficient, and P_k is its posterior covariance. After coefficient-domain denoising, the enhanced signal was reconstructed using multiresolution synthesis

$$\hat{x}(t) = \sum_{j \in \mathcal{J}} MRA_j(t)$$

with $\mathcal{J} = \{2, 3, 4\}$ for respiratory recordings, and $\mathcal{J} = \{4, 5\}$ for cardiac recording.

Average breath sensing profile and feature extraction. To prepare the signal for breathing cycle analysis, the denoised waveform was down-sampled to 1000 Hz. A full-wave rectified envelope was generated using a 200-ms moving window, and an energy-intensity signal was computed as the moving mean of the squared amplitude over a 0.5-s window. The energy trace was further smoothed using a Gaussian kernel ($\sigma = 50$ ms). Breathing-phase segmentation was performed by thresholding the smoothed energy curve; rising and falling crossings of the threshold yielded inhale and exhale start times, respectively. Individual inhalation and exhalation segments were extracted from the energy signal, time-aligned, and rescaled via temporal warping to match their average duration. For each subject, all inhale segments were warped to the mean inhalation duration, interpolated onto a shared time axis, and averaged to obtain a representative inhalation energy-density profile. The same procedure was applied to exhalation segments using their mean duration, and the exhalation profile was time-shifted to start immediately after the mean inhale interval, forming a continuous inhalation–exhalation template. Both inhalation and exhalation mean profiles were further modeled using a truncated Fourier series composed of sin- and cos-basis functions (six harmonics), enabling compact parametric representation suitable for real-time actuator control (supplementary note 3). The resulting time-normalized profiles were ultimately used as reference patterns for generating breathing-synchronized vibrotactile actuation.

Average cardiac sensing profile and feature extraction. The denoised cardiac signal was down-sampled to 1000 Hz for further analysis. A full-wave rectified envelope was computed and smoothed using a moving average of squared amplitude (0.1 s window) followed by Gaussian convolution ($\sigma = 50$ ms) to obtain an energy-intensity trace. Heartbeat segments were identified by thresholding this smoothed energy curve. Rising and falling crossings defined the start and end of each beat. Within each beat, local energy peaks were detected with minimum prominence and distance constraints, and the first two prominent peaks were labeled as S1 and S2, respectively. For each valid beat, S1 and S2 occurrence times and the S1–S2 interval were extracted, and across beats the mean S1–S2 duration, beat-to-beat cycle intervals (successive S1–S1), derived heart rate (beats per minute), and S2-to-next-S1 gaps were computed. To construct a representative heartbeat profile, each beat's smoothed energy waveform was resampled either onto a normalized phase axis (0–1 over the beat duration) or onto a common real-time axis (from 0 to the maximum observed beat duration). These warped waveforms were stacked and averaged to yield an ensemble mean heartbeat energy profile, with optional variability bands (mean $\pm$ SD or interquartile range). The resulting normalized cardiac energy template was used as a reference profile for designing heartbeat-synchronized vibrotactile actuation patterns.

Actuator control pattern. All vibrotactile patterns in the firmware were implemented as amplitude- and frequency-modulated carrier signals of the form

$$x(t) \approx A(t)\sin(2\pi f_{\text{carrier}}(t)t)$$

where $x(t)$ denotes the pulse width modulation (PWM)-modulated coil-drive signal, $A(t)$ is mapped directly to the PWM duty cycle (0 to 1), and $f_{carrier}(t)$ corresponds to the PWM switching frequency. Depending on the selected actuation mode, $A(t)$ and $f_{carrier}(t)$ were configured to reproduce different types of waveform modes (supplementary note 2).

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Data and materials availability: All data needed to evaluate the conclusions are present in the paper and the Supplementary Materials. Additional data and materials related to this paper may be requested from W.-H.Y.

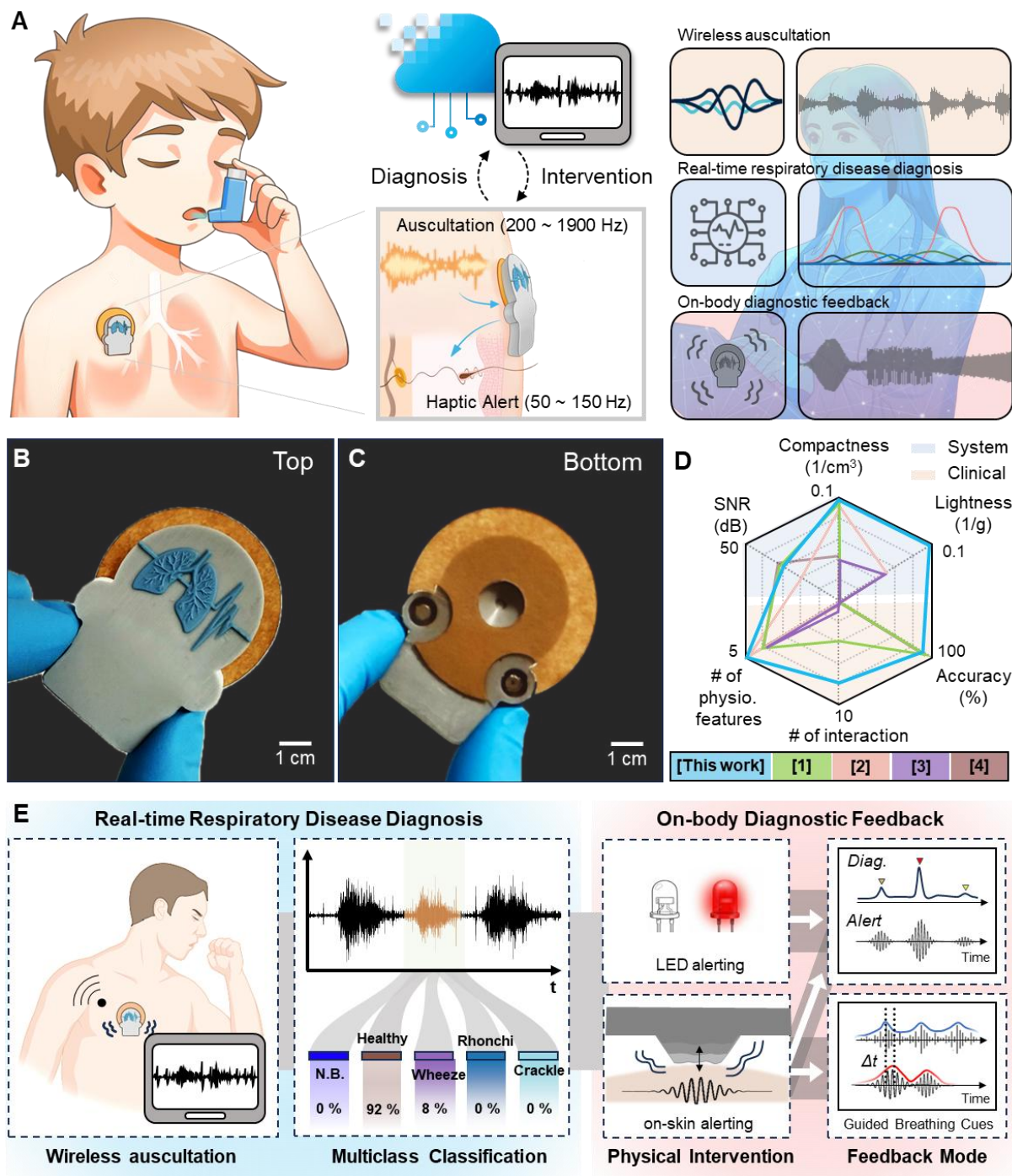


Fig. 1. On-body sensing-actuation integrated wearable system enabling real-time respiratory abnormality detection and on-body diagnostic feedback. (A) Conceptual illustration of the closed-loop wearable platform during use in a patient with respiratory disease. The skin-interfaced device continuously acquires cardiopulmonary acoustic signals through on-body auscultation and wirelessly streams the data to a portable hub. A real-time machine-learning algorithm analyzes the incoming signals to infer respiratory states, including clinically defined abnormal lung sounds, and generates corresponding diagnostic feedback. (B, C) Photographs of the device showing the top view, which includes the MEMS microphone and the integrated sensing-actuation circuit, and the bottom view, which incorporates dual electromagnetic coils and a reusable skin-safe adhesive layer. (D) Radar plot comparing key engineering performance and clinical relevance metrics including compactness, weight, signal-to-noise ratio (SNR), diagnostic accuracy, number of implemented

functions, and level of user interaction between the proposed system and reference works [1] *Jeong et al.* (33), [2] *Yoo et al.* (34), [3] *Spinelli et al.* (29), and [4] *Hui et al.* (35). **(E)** System-level workflow illustrating real-time respiratory disease assessment and on-body diagnostic feedback. Wireless auscultation data are processed using a multi-class machine-learning model to classify respiratory sounds into clinically relevant categories. Based on the inferred respiratory state, the system delivers immediate diagnostic feedback through multimodal on-body cues, including visual LED alerts and vibrotactile notifications, as well as guided-breathing cues that support patient awareness and respiratory self-management.

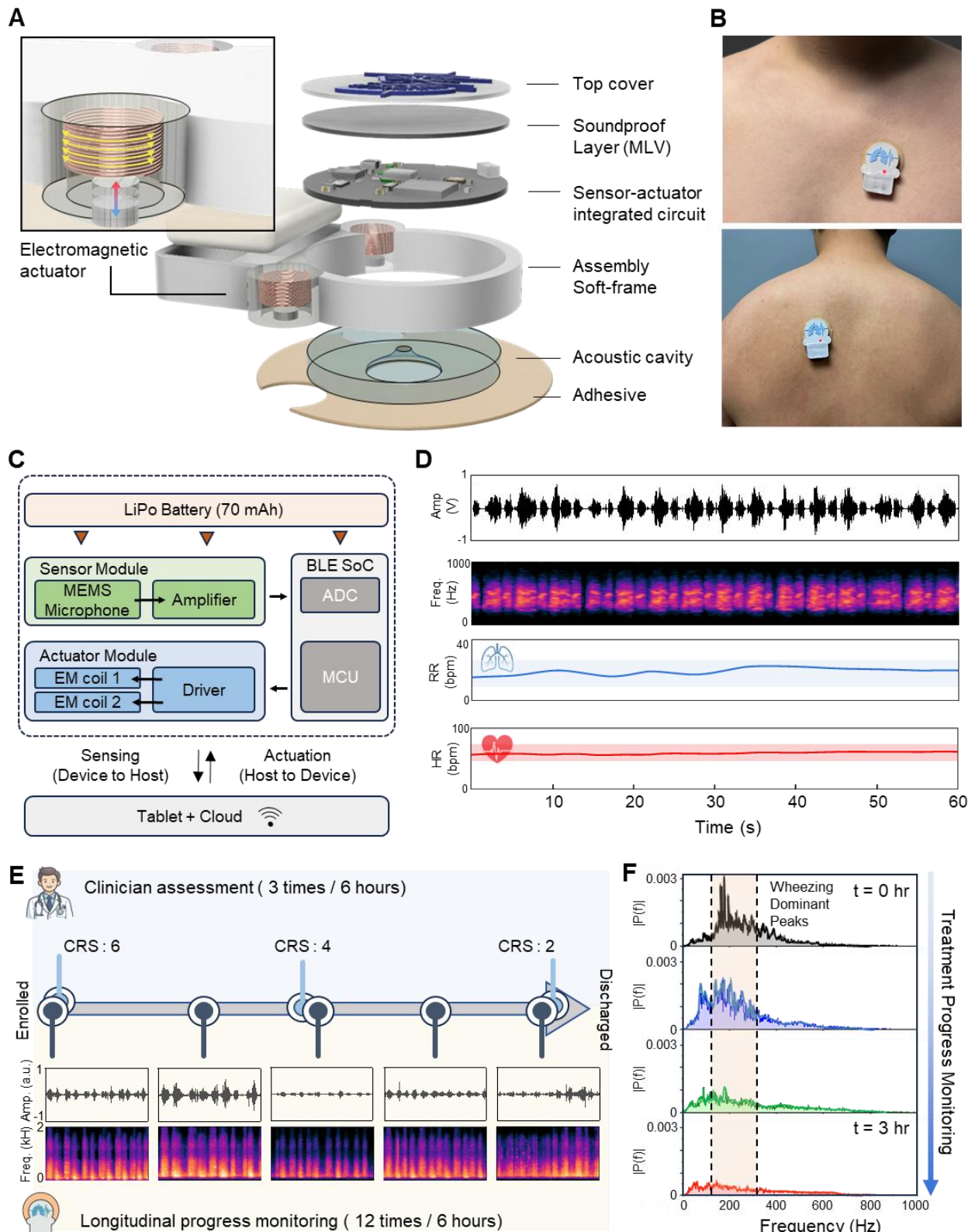


Fig. 2. Design, clinical implementation, and signal acquisition of the wearable system. (A) Exploded view of the multilayer device architecture. (B) Photographs of the device worn on the anterior (chest) and posterior (back) surfaces of the body. (C) System-level schematic of the sensing-actuation-based interactive platform. (D) Representative output of the signal-processing workflow, including the raw waveform (top), time-frequency spectrogram (second), extracted breathing envelope and breathing rate (third), and temporal energy distribution (bottom), enabling cycle-resolved characterization of respiratory behavior. (E) Clinical timeline and continuous

893 respiratory monitoring during hospitalization. Pediatric asthma participants received intermittent
894 clinician assessments using the Clinical Respiratory Score (CRS) and continuous longitudinal
895 respiratory sound acquisition by the device during the admission-to-discharge period. **(F)**
896 Longitudinal monitoring of asthma participants over a 3-hour treatment period. STFT analysis
897 shows that strong, narrow-band wheeze-related peaks present at enrollment gradually diminish as
898 treatment progresses, eventually transitioning to a broadband spectral distribution characteristic of
899 normal breathing.

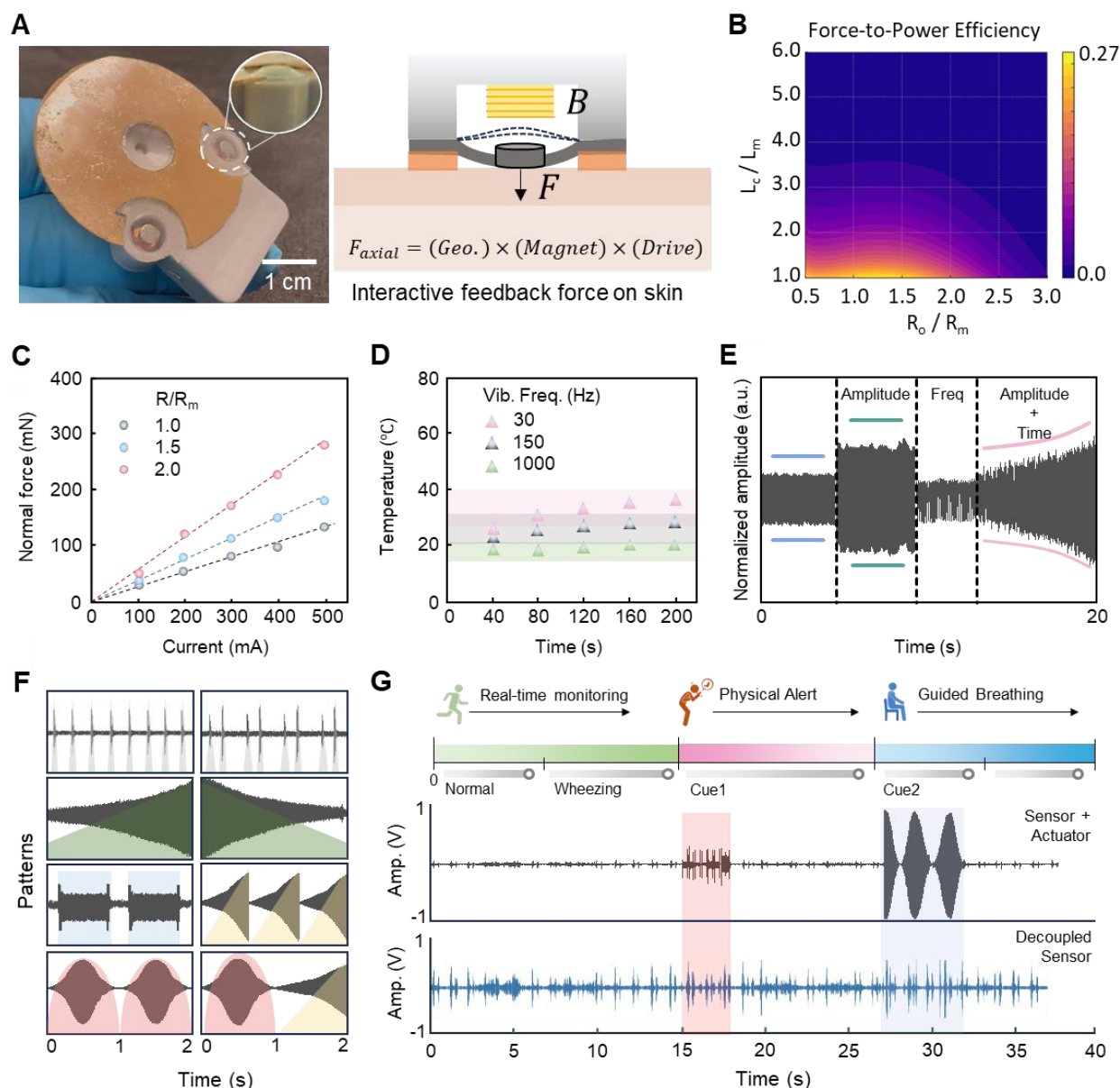


Fig. 3. Characterization and real-time integration of the sensing-actuation module. (A) A photograph and schematic illustration of the integrated actuator module. The actuation principle is governed by axial force generation that depends on the coil-magnet geometry, magnetic properties, and drive current. (B) Simulated normalized force-to-efficiency generated by the actuator under excitation. (C) Linear correlation between input drive current and measured actuation force. (D) Temperature measurements of the actuator under different operating frequencies. (E) Measured vibration amplitudes at different actuation frequencies, showing consistent and frequency-tunable output. (F) Representative examples of tactile patterns with varying amplitude, frequency, and duration. (G) Simultaneous sensing and actuation during real-time measurement. The upper traces show the actuator driving waveform and real-time sensor output recorded concurrently. The middle panel displays the separated components, distinguishing sensor-originated acoustic signals from actuation-induced vibrations through signal decoupling.

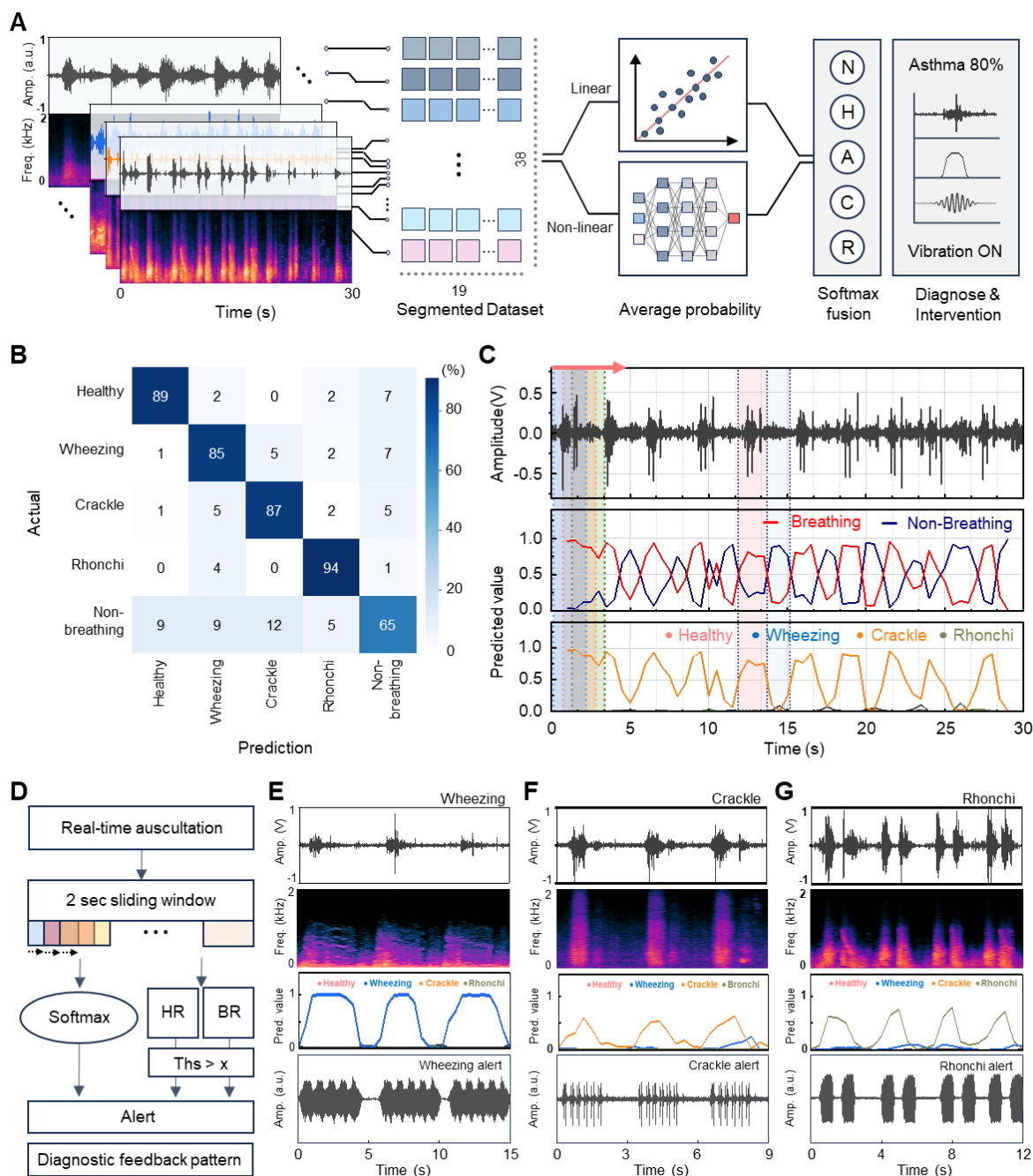


Fig. 4. Machine-learning-based multi-class classification and real-time feedback framework. (A) Schematic representation of the multi-class classification model and respiratory state inference. (B) Confusion matrix of the multi-class classification model performance. (C) Real-time machine-learning feedback flow showing synchronized waveform, spectrogram, and feedback activation timeline within a diagnostic window. (D) System-level workflow linking real-time machine-learning diagnostics to tactile feedback generation. (E to G) Abnormal respiratory classifications or deviations in breathing or heart rate beyond predefined thresholds trigger immediate alerting and selection of the corresponding actuation pattern. Representative examples showing continuous waveform, spectrogram, multi-class probability distribution, and triggered feedback for subjects exhibiting (E) wheezing, (F) crackle, and (G) rhonchi.

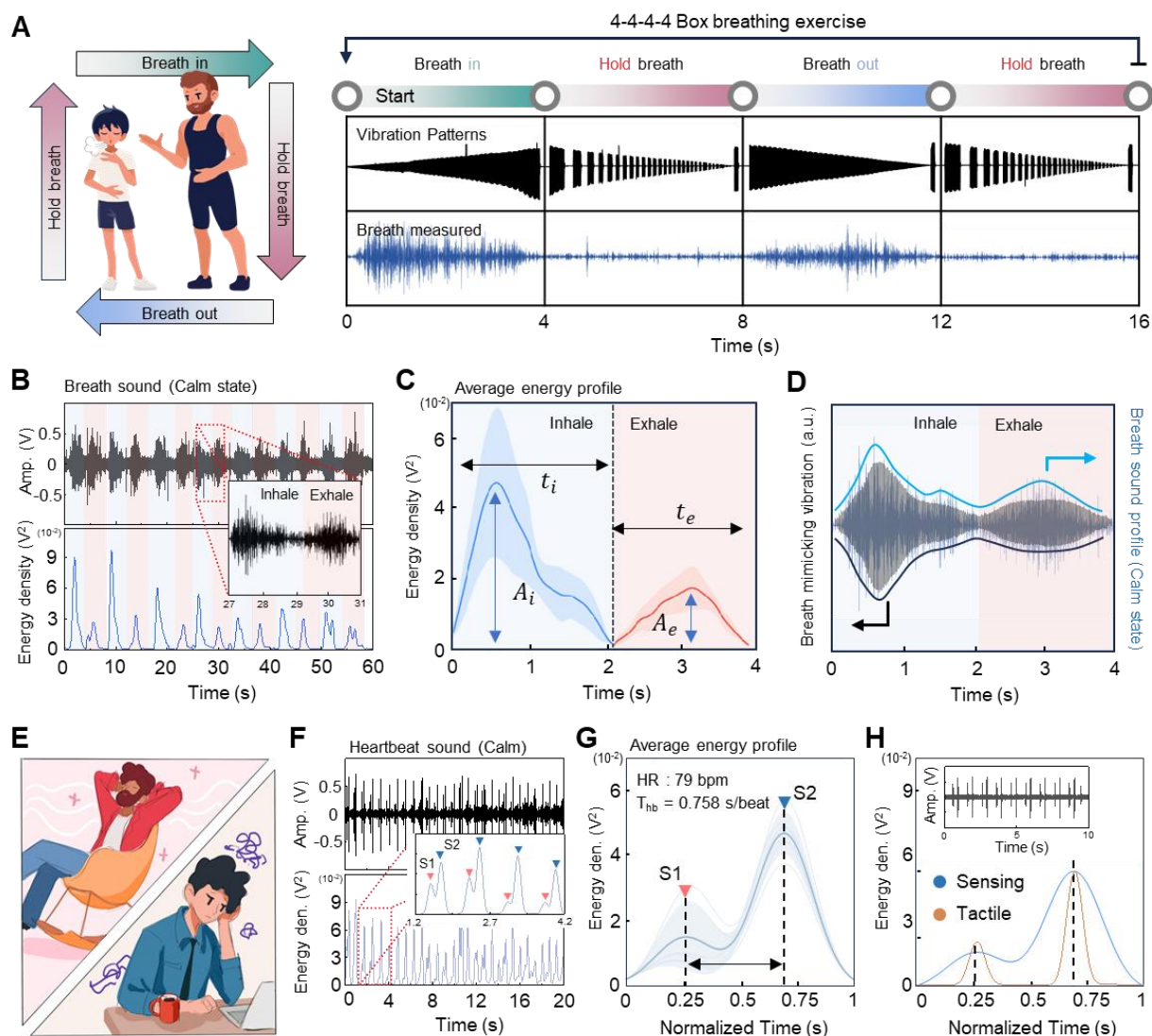


Fig. 5. Personalized guided-breathing and vibrotactile patterns mimicking individual breathing and heart sound feedback. (A) Guided tactile pattern for box breathing exercise. Vibrotactile cues represent the four sequential phases of the breathing cycle (inhale–hold–exhale–hold). The vibration envelope and temporal modulation encode the duration and transition of each phase, enabling users to follow the breathing rhythm through tactile guidance. (B) Example of a calm-state breathing pattern measured for tactile pattern calibration. The upper trace shows the raw breath sound, while the lower trace presents the corresponding energy-density plot. (C) Energy-density statistics for each breathing cycle extracted from the calm-state breathing measurement. (D) Comparison between the tactile pattern designed to mimic the calm breathing (gray shading) and the actual breath-sound profile (blue line). (E) Illustration of the synchronized heartbeat-feedback mode designed to support stress relief. (F) Cardiac sound recorded during the calm state for tactile pattern calibration, along with its associated energy density. The inset shows a magnified energy-density profile of the 1.2 - 4.2 s window. (G) Energy-density statistics of individual calm-state cardiac cycles (S1 and S2) together with the extracted heart rate (HR, bpm). (H) Spike-type tactile pattern generated to mimic the cardiac rhythm based on extracted features, and comparison of its temporal characteristics with those of the actual heartbeat.